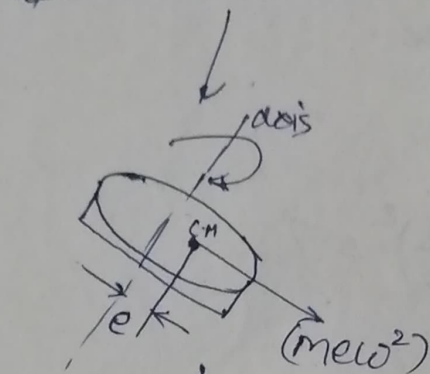


★ Balancing ★

Vibration causing Unbalance forces in a system

1. Rotational Unbalance

Const in Mag.
but changing
in dirⁿ



Balancing

$$\left[\begin{array}{l} \sum \vec{F} = 0 \\ \sum \vec{M} = 0 \end{array} \right]$$

when all the rotating masses are in same plane

① Static Balancing

In this balancing we balance only forces

Here moments are automatically balanced

2. Reciprocating Unbalancing

→ Const in dirⁿ
but changing
in Magnitude

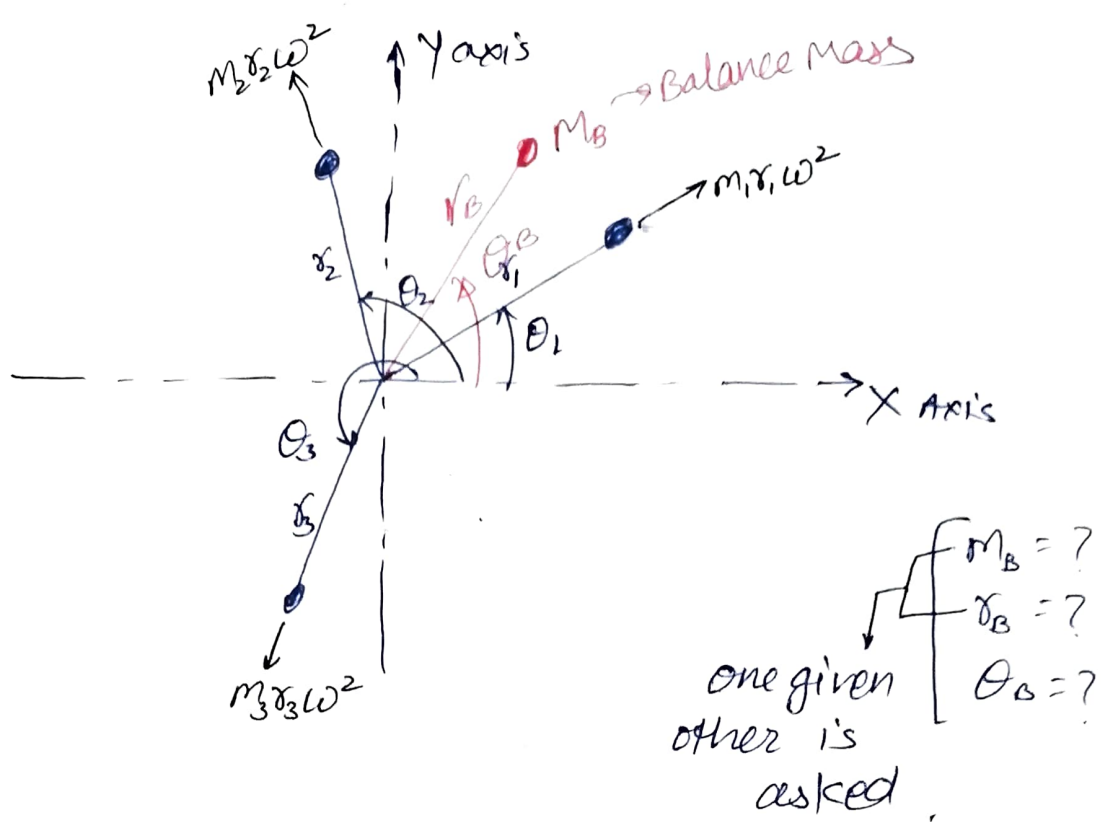
② Dynamic Balancing
[When all the masses are not rotating in same plane]

Here we balance both forces + Moments.

★ Static Balancing —

"All the masses are rotating in the same plane."

[Balancing is always independent of speed]



After Balancing

$$\Sigma \vec{F} = 0 \Rightarrow F_x = 0, F_y = 0$$

$$\underline{F_x = 0} \therefore (m_1 r_1 \omega^2) \cos \theta_1 + (m_2 r_2 \omega^2) \cos \theta_2 + (m_3 r_3 \omega^2) \cos \theta_3 + (m_B r_B \omega^2) \cos \theta_B = 0$$

$$\Rightarrow (m_B r_B) \cos \theta_B = -\Sigma m r \cos \theta \quad \text{--- (1)}$$

$$\underline{F_y = 0} \quad (m_B r_B) \sin \theta_B = -\Sigma m r \sin \theta \quad \text{--- (2)}$$

$$\sqrt{(1)^2 + (2)^2} \quad m_B r_B = \sqrt{(-\Sigma m r \cos \theta)^2 + (-\Sigma m r \sin \theta)^2}$$

$$\textcircled{2}/\textcircled{1} \quad \tan \theta_B = \frac{-\Sigma m r \sin \theta}{-\Sigma m r \cos \theta}$$

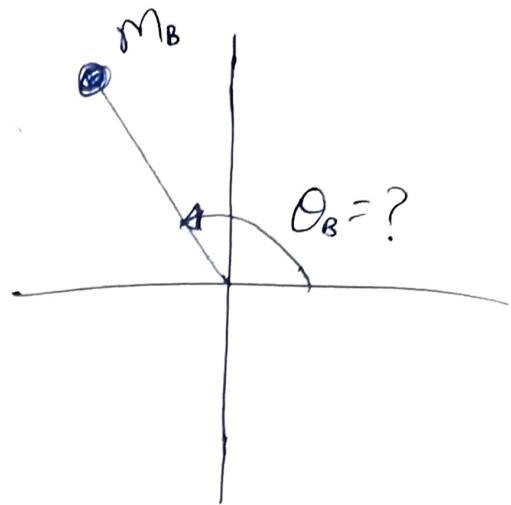
Note - The final signs of Numerator + Denominator will decide the Quadrant of Balance Mass.

For eqm.

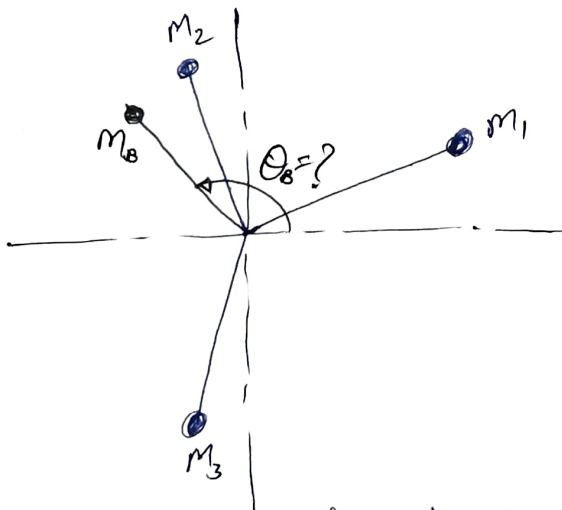
$$\sum m r \cos \theta = +4$$

$$\sum m r \sin \theta = -3$$

$$\begin{aligned} \tan \theta_B &= \frac{-\sum m r \sin \theta}{-\sum m r \cos \theta} \\ &= \frac{-(-3)}{-(+4)} \\ &= \frac{+3}{-4} = -\left(\frac{3}{4}\right) \end{aligned}$$



Graphical



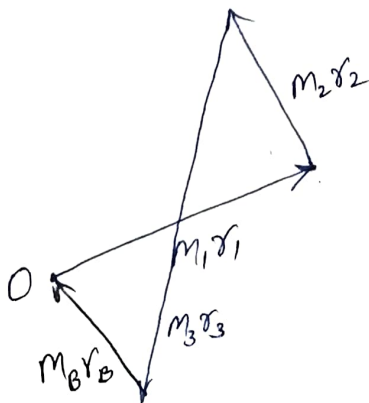
Configuration Diagram.

No scaling required

only correct angle should be plotted.

Force Polygon

Scale $\rightarrow$



(F/w²)

$\left. \begin{matrix} m_1 r_1 \\ m_2 r_2 \\ m_3 r_3 \end{matrix} \right\} \text{Known}$

$m_B r_B \text{] Unknown}$

Prob Three masses are rotating in a same plane.

$$m_1 r_1 = 2$$

$$m_2 r_2 = 4$$

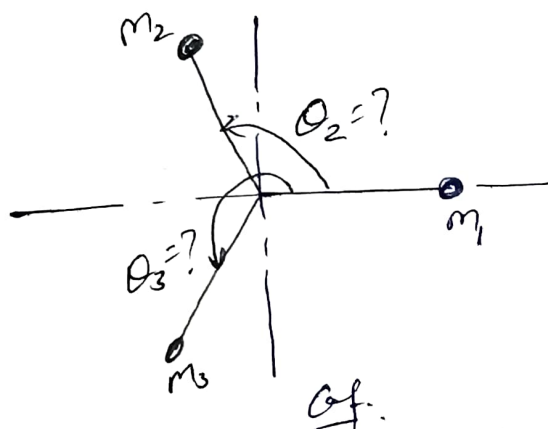
$$m_3 r_3 = 5$$

What should be the orientation of mass
② & ③ w.r.t. mass ① in order to have
complete static balance.

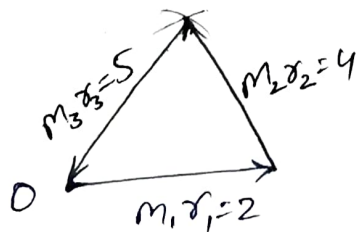
Solⁿ

$$\left. \begin{array}{l} \theta_2 = ? \\ \theta_3 = ? \end{array} \right\}$$

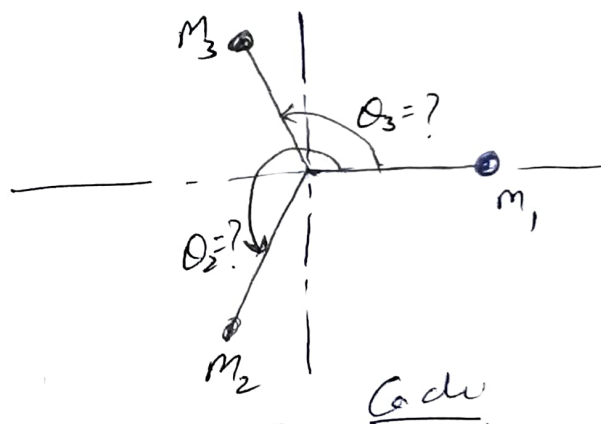
Graphical



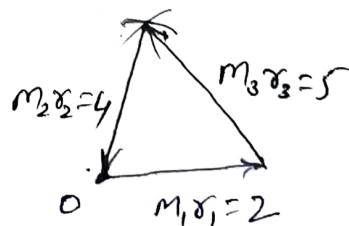
Force Polygon
scale



Graphical



Force Polygon
scale



$$(\theta_2, \theta_3)$$

$$(\angle, \angle) \rightarrow$$

$$(\angle, \angle) \rightarrow$$

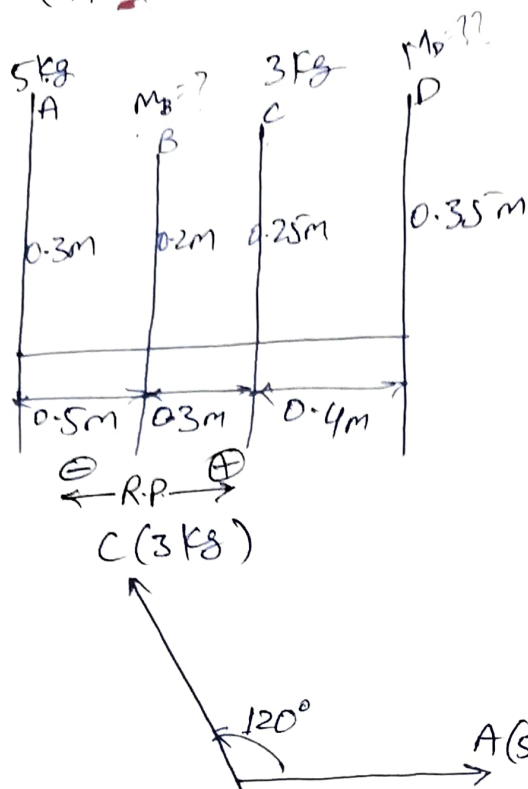


★ Dynamic Balancing

②

(All the masses are not rotating in a same plane)

pb:-

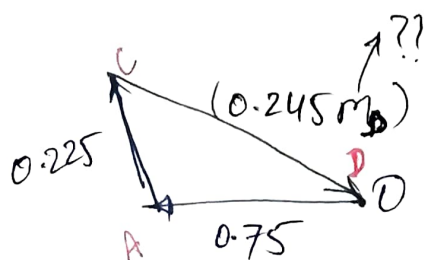


Find the masses in the planes of B + D along with their positions in order to have complete balancing:

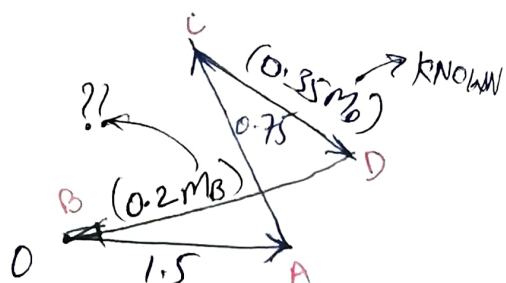
$$\left. \begin{array}{l} m_B = ? \\ m_D = ? \\ \theta_B = ? \\ \theta_D = ? \end{array} \right\}$$

Planes	m	z	m.z	Distance from R.P. (l)	(m.z.l)
A	5	0.3	1.5	-0.5	-0.75
B(R.P.)	m_B	0.2	$(0.2m_B)$	0	0
C	3	0.25	0.75	0.3	0.225
D	m_D	0.35	$(0.35m_D)$	0.7	$(0.245m_D)$

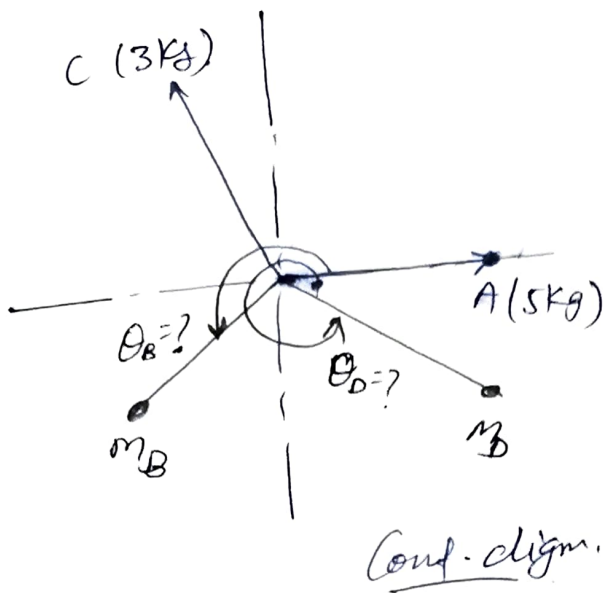
Moment Polygon
Scale



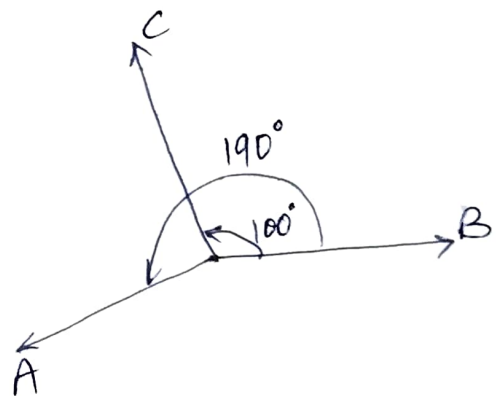
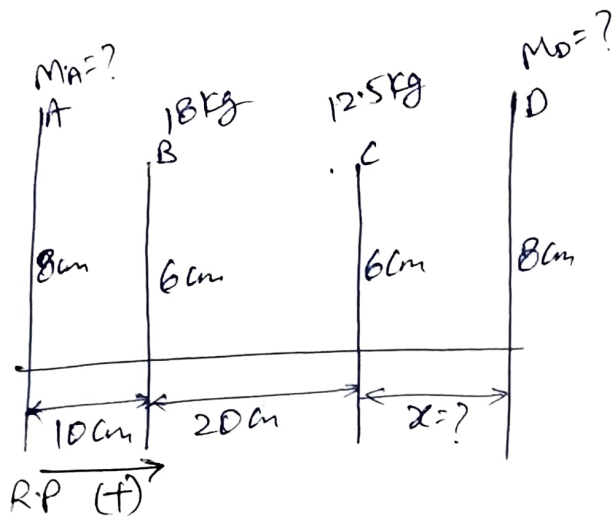
Force Polygon
Scale



Graphical.

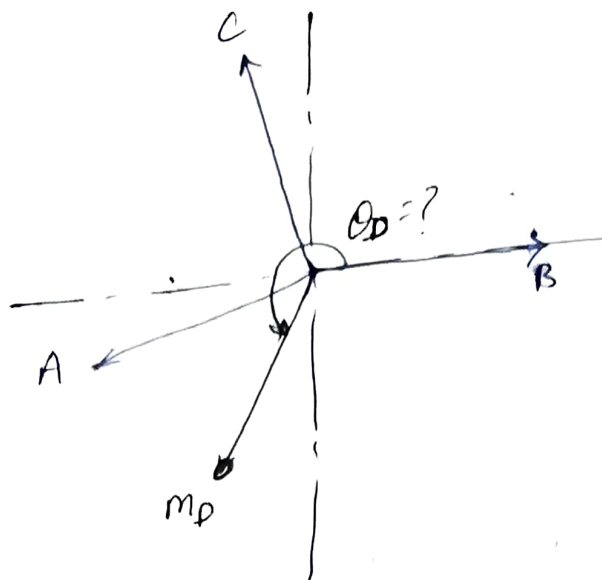


WB
Pb-2

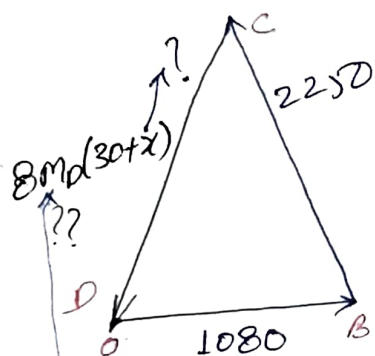


Planes	m	z	m.z	Distance from R.P. (cm)	(m.z.l.)	
A(R.P.)	m_A	8	$8m_A$	0	0	
B	18	6	108	10	1080	Scale (mm)
C	12.5	6	75	30	2250	
D	m_D	8	$8m_D$	$(30+x)$	$8m_D(30+x)$	

Graphical

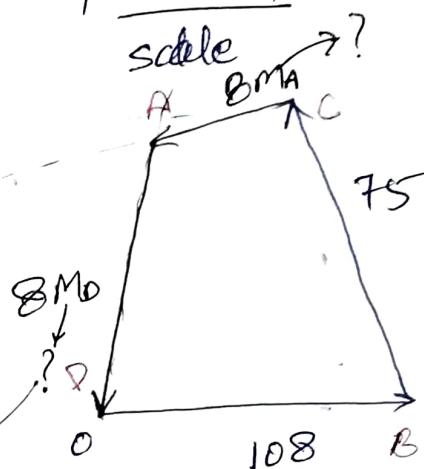


Moment Polygon scale



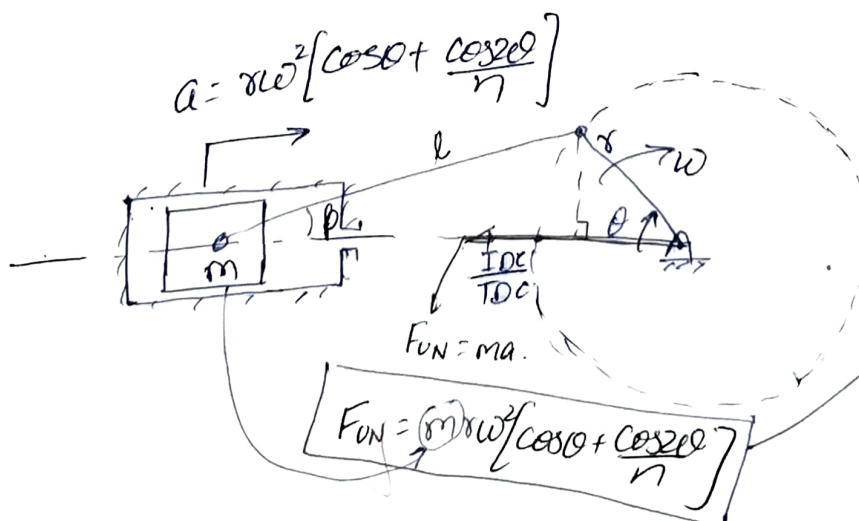
Force Polygon

scale



Balancing of Reciprocating Masses

$m \Rightarrow$ mass of reciprocating parts of engine.

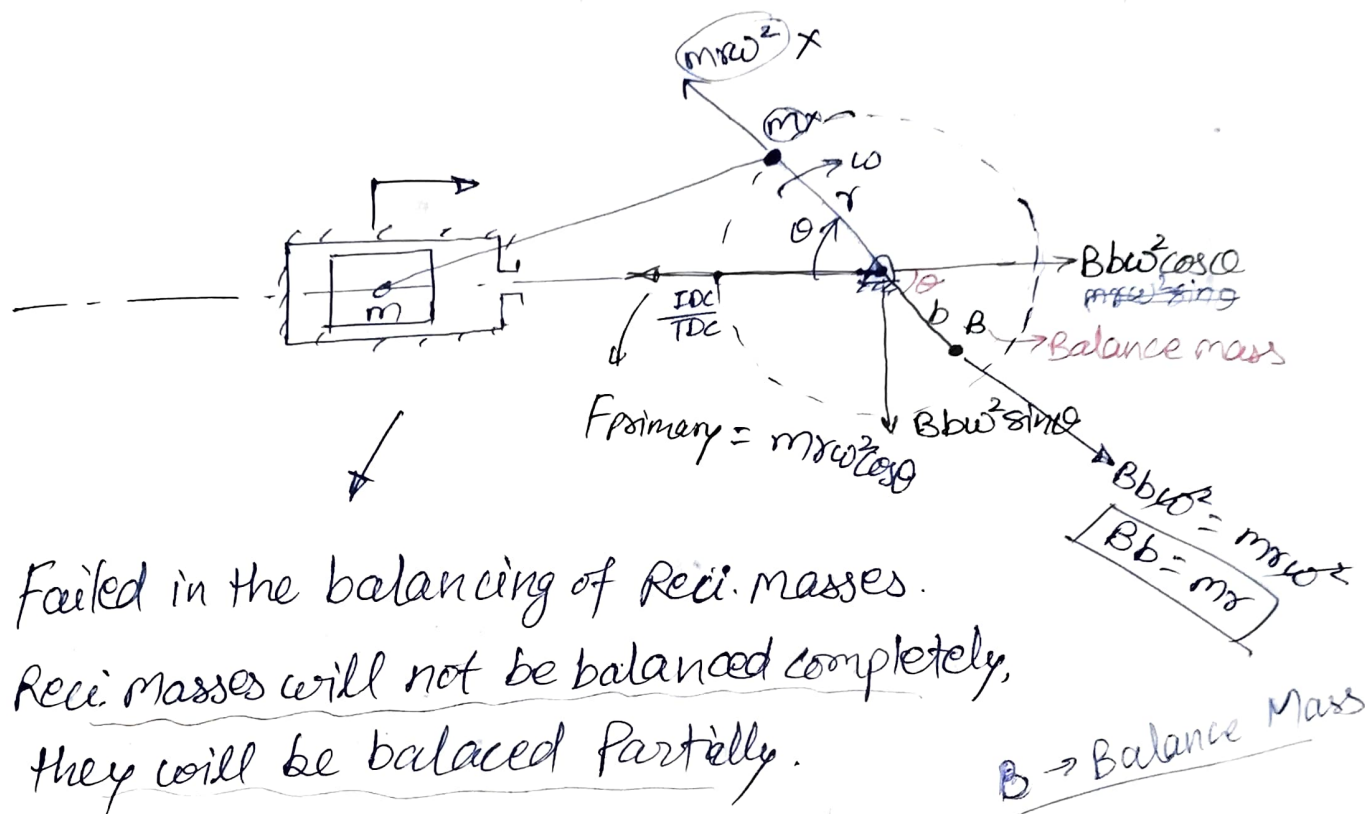


This force is always const in dirⁿ but every moment change in magnitude

$$F_{un} = \underbrace{mr\omega^2 \cos\theta}_{F_{un}(\text{primary})} + \underbrace{mr\omega^2 \cos 2\theta}_n$$

$F_{un}(\text{primary}) \gg \gg F_{un}(\text{secondary})$

* Primary Balancing of Reciprocating Masses -



Partial Balancing of Rec. Masses -

Let $c \rightarrow$ Fraction of Rec. mass to be balanced

$$0 < c < 1$$

$$B.b = c.m.r < m.r$$

After Partial Balancing

$$F_{un}(\text{Along line of stroke}) = mr\omega^2 \cos\theta - Bb\omega^2 \cos\theta$$

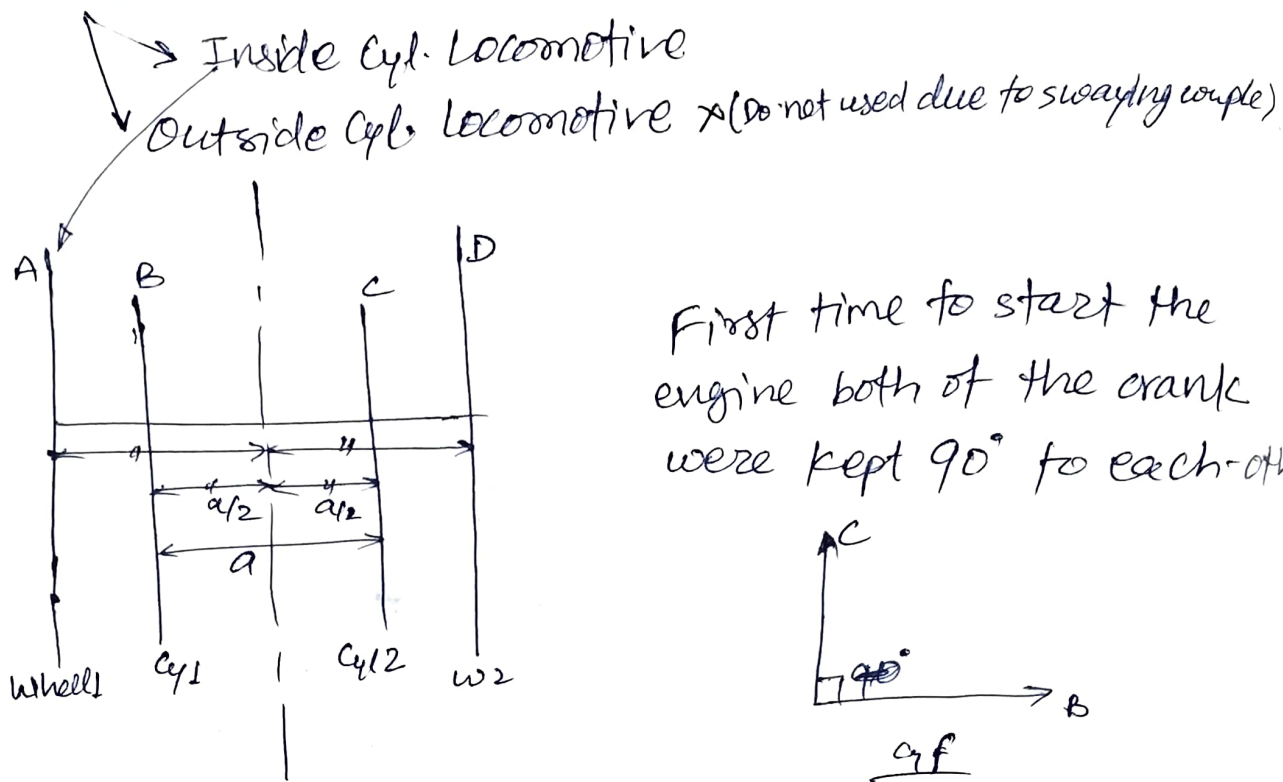
$$= (1-c)mr\omega^2 \cos\theta$$

$$F_{un}(\perp \text{ to LOS}) = Bb\omega^2 \sin\theta$$

$$= cmr\omega^2 \sin\theta$$

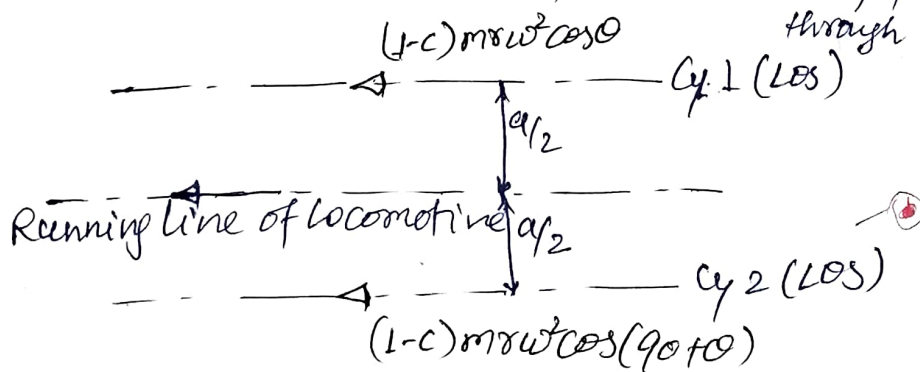
✓ Success in Running engines for longer period)

Two-Cyl. - Locomotive -



Effects of Partial Balancing on Two-Cyl. Locomotive

1. Variation in Tractive force - used to generate motion b/w body & a tangential surface through the use of dry friction



$$\text{Tractive force} = (1-c)mr\omega^2\cos\theta + (1-c)mr\omega^2\cos(90+\theta)$$

$$= (1-c)mr\omega^2[\cos\theta - \sin\theta] \quad (1)$$

For this to be max:

$$\frac{d}{d\theta}(\cos\theta - \sin\theta) = 0 \Rightarrow \tan\theta = -1, \theta = 135^\circ, 315^\circ$$

* Variation in Tractive force = $\pm\sqrt{2}(1-c)mr\omega^2$

2. Variation in Swaying Couple — tend to sway alternatively in cw ccw

$$\begin{aligned}\text{Swaying couple} &= (1-c) m r \omega^2 \cos \theta \cdot \frac{a}{2} - (1-c) m r \omega^2 \cos(\theta + \theta) \frac{a}{2} \\ &= \frac{a}{2} (1-c) m r \omega^2 [\cos \theta + \sin \theta]\end{aligned}$$

for this to be max -

$$\frac{d}{d\theta} (\cos \theta + \sin \theta) = 0$$

$$\Rightarrow \tan \theta = 1 \Rightarrow \theta = 45^\circ, 225^\circ$$

$$\# \boxed{\text{Variation in Swaying Couple} = \pm \frac{a}{2} (1-c) m r \omega^2}$$

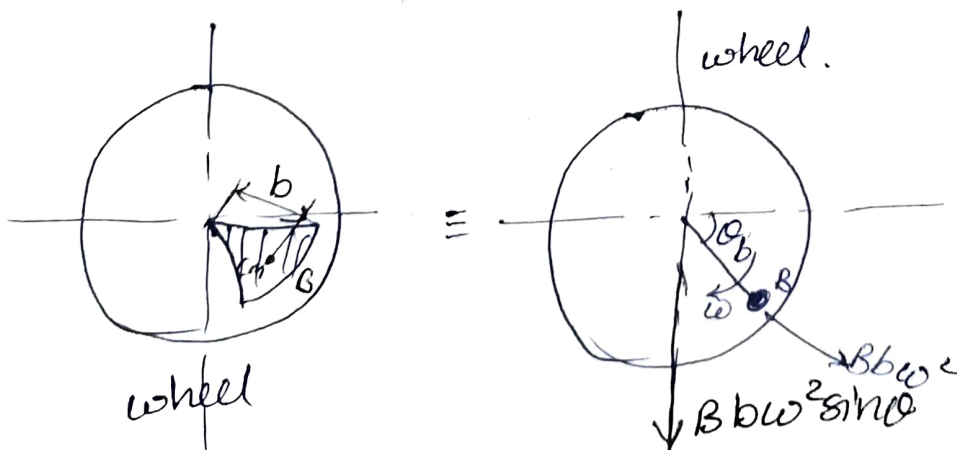
3. Hammer Blow : - $B b \omega^2$

In the Partial balancing of Rec. Masses.

⇒ Balance mass can't be attached reverse to the crank

⇒ Balance masses are attached to the wheels.

$$\underline{B \cdot b \neq c \cdot m r}$$



$$(B b \omega^2 \sin \theta)_{\max} = \frac{B b \omega^2}{\text{}} \rightarrow \text{Hammer Blow.}$$

$$(B b \omega^2) \leq \text{Per wheel static load.}$$

↳ Speed $\rightarrow$ limited

$$\text{Hammer Blow} = B \omega^2$$

Balance mass responsible for the balancing of Reciprocating of cyl. only.

Pb.

Two cyl. locomotive -

$$\text{Rot. mass/cyl} = 200 \text{ kg}$$

$$\text{Reci mass/cyl} = 400 \text{ kg}$$

$$\text{Stroke} = 0.8 \text{ m}$$

$$\Rightarrow r = 0.4 \text{ m}$$

$$\text{Cyl. Planes gap} = 1.0 \text{ m}$$

$$\text{wheel planes gap} = 1.5 \text{ m}$$

$$N = 300 \text{ rpm}$$

Both of cranks are 90° to each other

Balance masses are attached on the wheels at a distance of 0.3 m from the wheel centres in order to balance

$$\left(\text{Rot.} + \frac{3}{4} \text{ Reci} \right)$$

find -

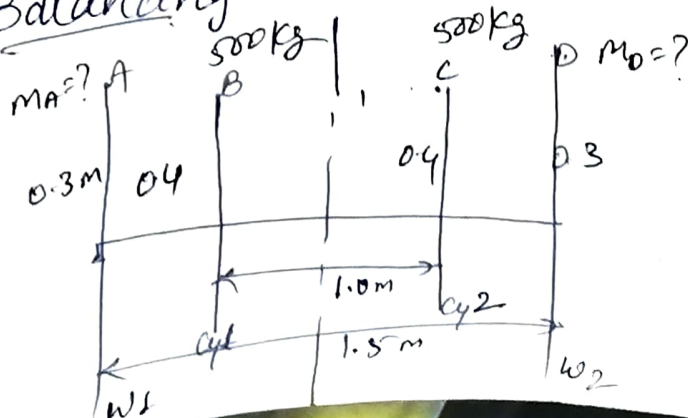
1) Balance masses along with their positions in order to have balancing

2) Also find Hammer blow

Solⁿ

$$\begin{aligned} \text{Rot.} + \frac{3}{4} \text{ Reci} &= 200 + \frac{3}{4} \times 400 \\ &= 200 + 300 \\ &= 500 \text{ kg/cyl.} \end{aligned}$$

(i) Balancing -



$$\begin{aligned} M_A &= ? \\ M_D &= ? \\ \theta_A &= ? \\ \theta_D &= ? \end{aligned}$$

Same for B and C



ii) Hammer Blow = $B b \omega^2$

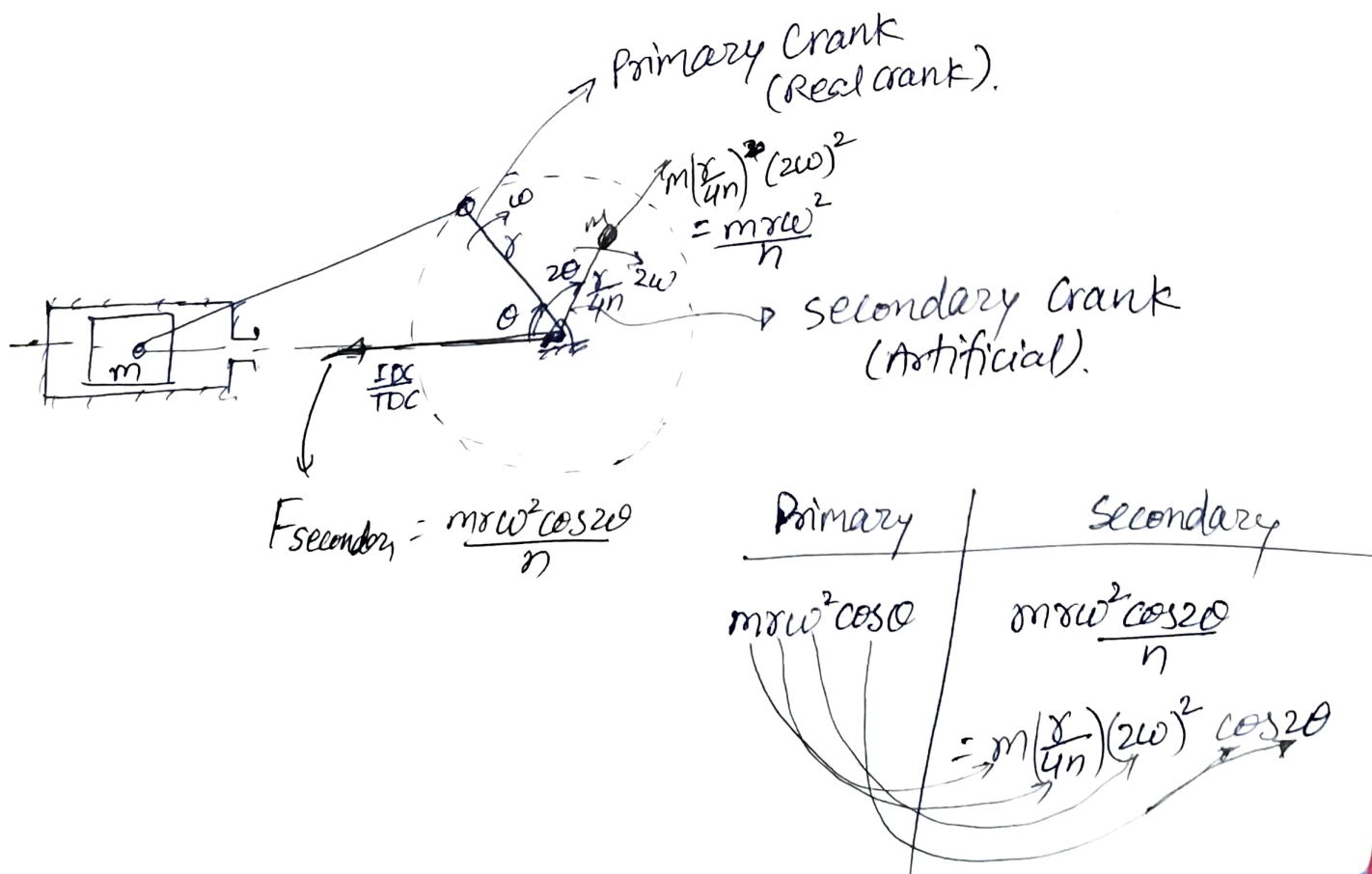
$$\begin{array}{lcl} 500 \text{ kg} & \longrightarrow & 80 \text{ kg} \\ 300 \text{ kg} & \longrightarrow & \left(\frac{80}{500} \times 300 \text{ kg} \right) \\ \text{Rec. mass} & & B \end{array}$$

$B \rightarrow$ Balance mass

Introduction of the Coupling Rod in Two-Cyl. Locomotives

" Coupling rod was basically introduced to decrease the amount of Hammer Blow by decreasing the balance mass requirement for the reciprocating balancing through splitting the reci. balancing b/w the driving & trailing wheels. This was the development from passenger locomotives to the express locomotives in which 2-2 wheels were coupled, to the super fast locomotives in which 3-3 wheels were coupled.

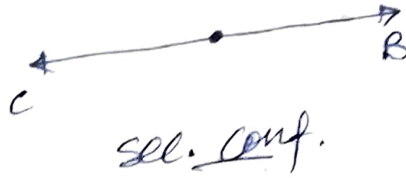
★ Secondary Balancing :-



For Exm

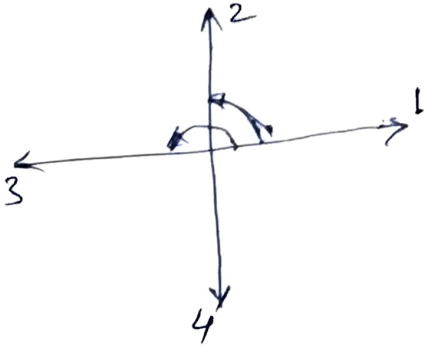


Primary Conf.



sec. Conf.

4-Cyl. Inline Engine

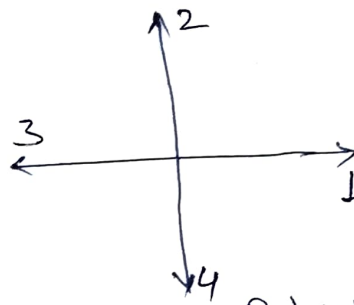
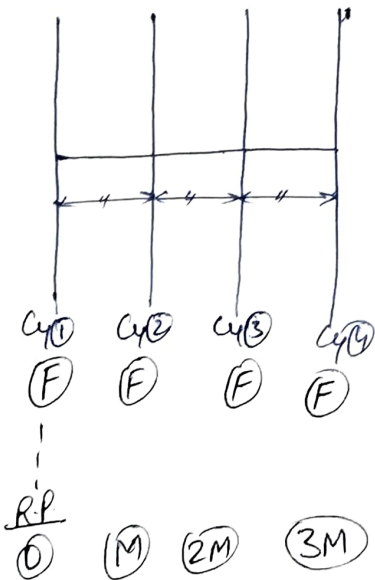


Primary Conf.



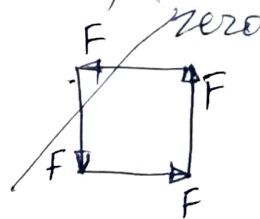
secondary Conf.

Role of the Firing Order in a Multi Cylinder Inline Engine in Balancing

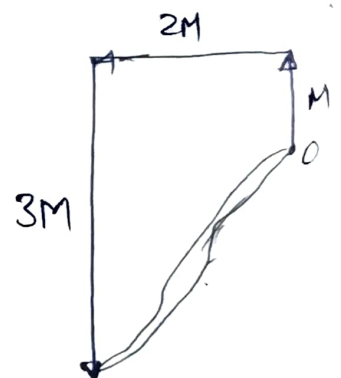


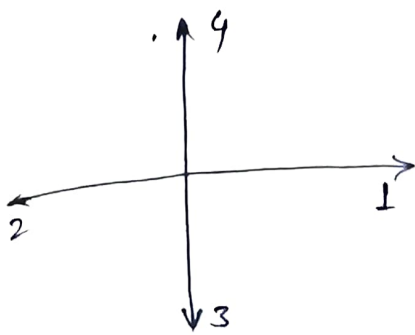
Primary Conf.

Primary force Polygon



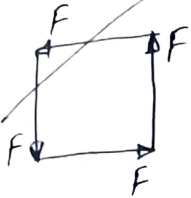
Primary Moment Polygon



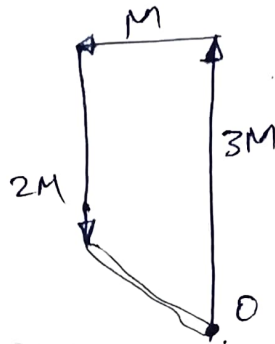


Prm. Conf.

PFP: - zero



PMP: -

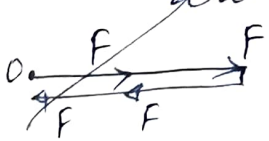


The Real firing order of 4-Cyl. Inline engine.

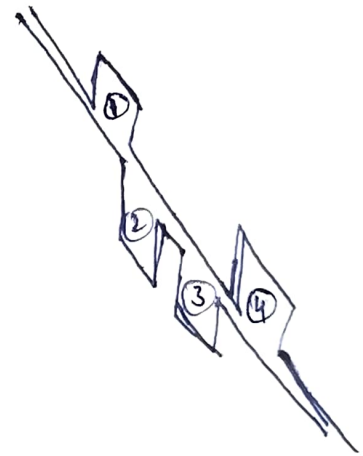
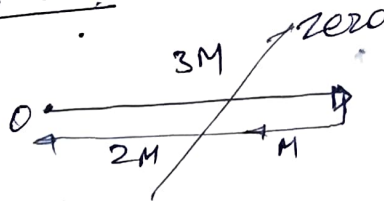


Prm. Conf.

PFP: - zero



PMP: -



* 4 Cyl. Inline Engine

↳ Not completely balanced.

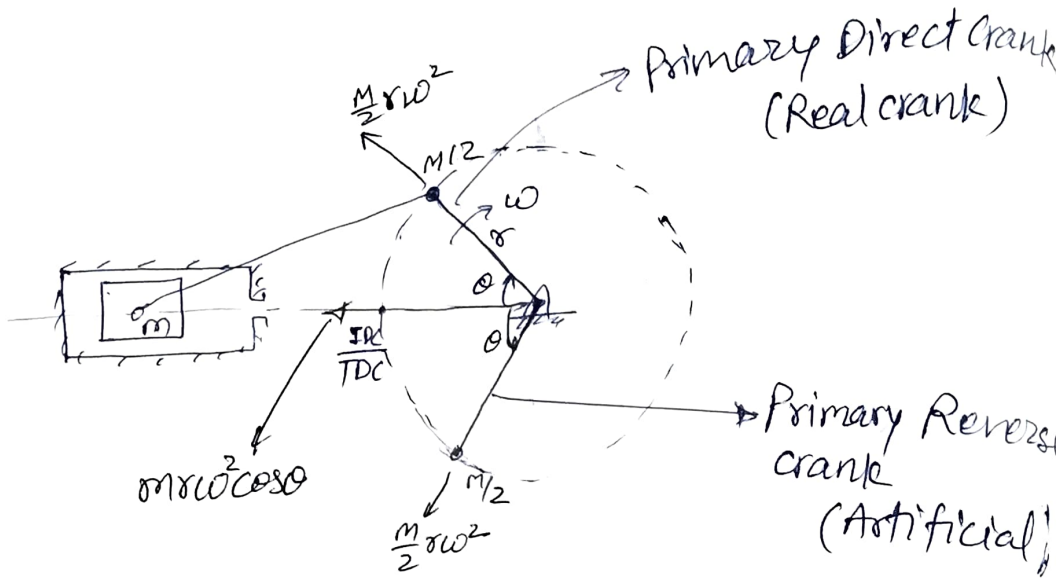
* Six-Cyl. Inline Engine

↳ completely balanced.

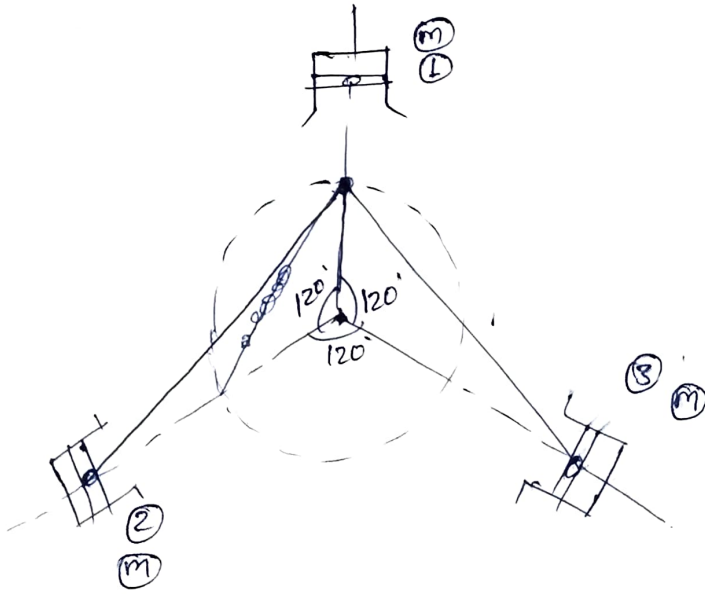
(1-4-6-5-2-3) → firing order.

* Direct and Reverse Crank Method :-

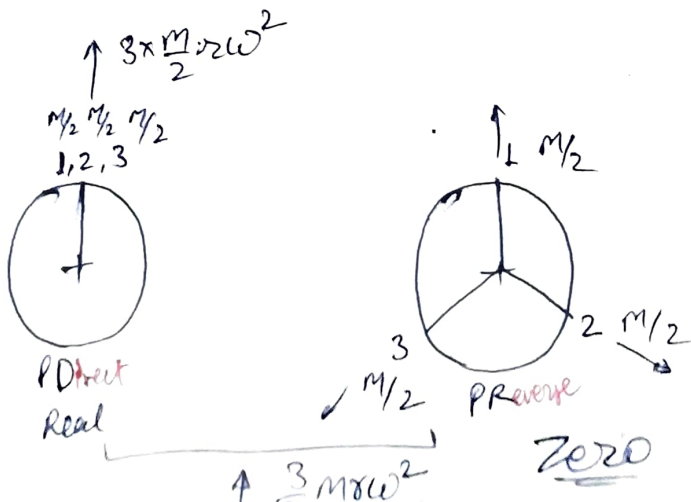
"This method is basically applied in multicylinder radial engines to get the magnitude & the dirⁿ of unbalanced primary and secondary forces."



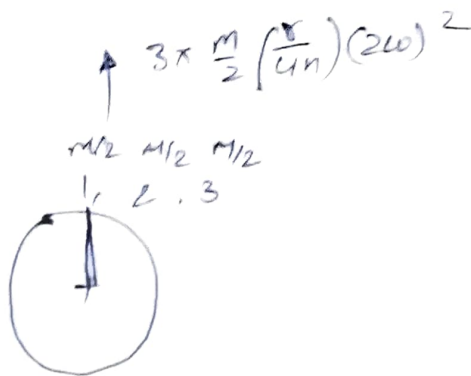
Three Cyl. RADIAL ENGINE



F_{primary} = ?



F_{secondary} = ?



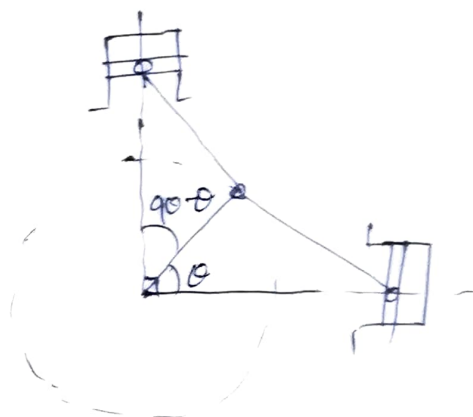
$$\uparrow \frac{3}{2} \frac{m r \omega^2}{n}$$

$$\uparrow \frac{3}{2} \frac{m r \omega^2}{n}$$

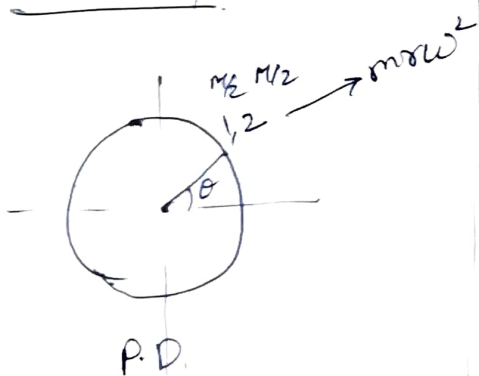
IES-2010
10 Marks

90° - V Engine

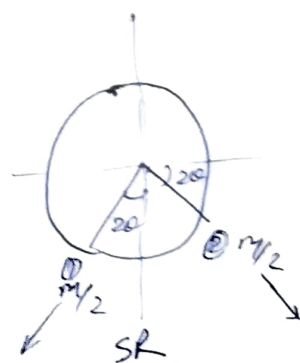
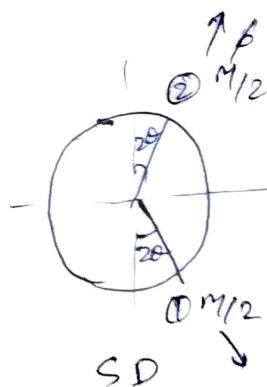
F_{primary} } Max?
 $F_{\text{secondary}}$ } Min?



Primary



Secondary

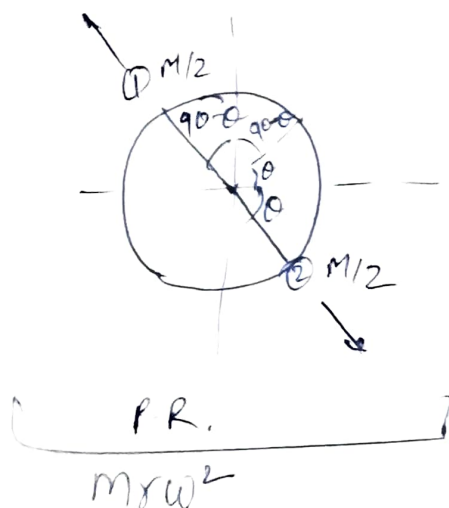


$$\rightarrow F_x = \frac{m}{2} \frac{r \omega^2}{n} [\cos 2\theta + \sin 2\theta + \cos 2\theta - \sin 2\theta]$$

$$= \frac{m r \omega^2 \cos 2\theta}{n}$$

$$\rightarrow F_y = \frac{m}{2} \frac{r \omega^2}{n} [\sin 2\theta - \cos 2\theta - \cos 2\theta + \sin 2\theta]$$

$$= - \frac{m r \omega^2 \cos 2\theta}{n}$$



$$F_{\text{secion}} = \sqrt{2} \frac{m r \omega^2 \cos 2\theta}{n}$$

max at $\theta = 0$
min at $\theta = 90^\circ$